IGCSE Math Extended 0580

Chapter 3:
Algebra and Graphs

Welcome to AnithUncommon's **Sample Resources!**

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In IGCSE Math Extended 0580, 3.1-3.5

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by (Mentor Name) to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in (Subject Name), join our nonprofit!

Gmail: anithuncommon
Instagram handle: @anithuncommon

This sample is made thanks to, Vihaan Amin

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

- **Construct and interpret** linear graphs from tables of values or given equations.
- **Calculate** the "steepness" or Gradient of any line using two fixed points.
- **Find the exact physical properties** of a line segment, including its total length and its center point.
- **Derive and manipulate** the general equation of a straight line in the form y = mx + c.

Reflective Questions to Consider:

Coordinate geometry is the visual language that connects equations to the physical world, allowing us to map out abstract rules onto a grid. By defining positions with exact values, we can calculate the precise behavior, steepness, and distance of any line. Mastering these coordinates transforms a blank graph into a map where every line follows a logical direction.

- Why is a horizontal line said to have a gradient of zero, while a vertical line's gradient is undefined? What does this tell us about the "rate of change"?
- How is calculating the length of a diagonal line on a grid just a version of Pythagoras' Theorem?

- If the equation of a line is its "name," how do the values of m and c change the way that line looks on a map?

Syllabus Outline:

<table>
<tr><th colspan="2">Unit 3: 3.1-3.5</th></tr>
<tr><td colspan="2">

Tip!

- **The x-value (horizontal) always comes before the y-value (vertical). When calculating gradients, the most common mistake is swapping the order, always keeping the y-values on top of the fraction. Think of the Gradient as "Rise over Run."**

Activities in Class:

- **Collaborative Virtual Whiteboard: Students and mentors use a shared digital board to solve complex problems.**
- **Solving a real world Systems of Equations "mystery". You are given a digital prompt with the costs of different meal deals and must find the price of individual items.**
- **Digital Drag-and-Drop: Sorting algebraic expressions into virtual categories: "Factorizable," "Difference of Two Squares," and "Prime Expressions."**

</td></tr>
<tr><td>

3.1 - Coordinates

- **The coordinate plane** is a way to describe location using two directions: **horizontal (x) and vertical (y)**. Every point is defined by how far you move in each direction.
- The **origin (0, 0)** is the **starting point** because both values are zero, meaning no movement. All other points are measured relative to this point.

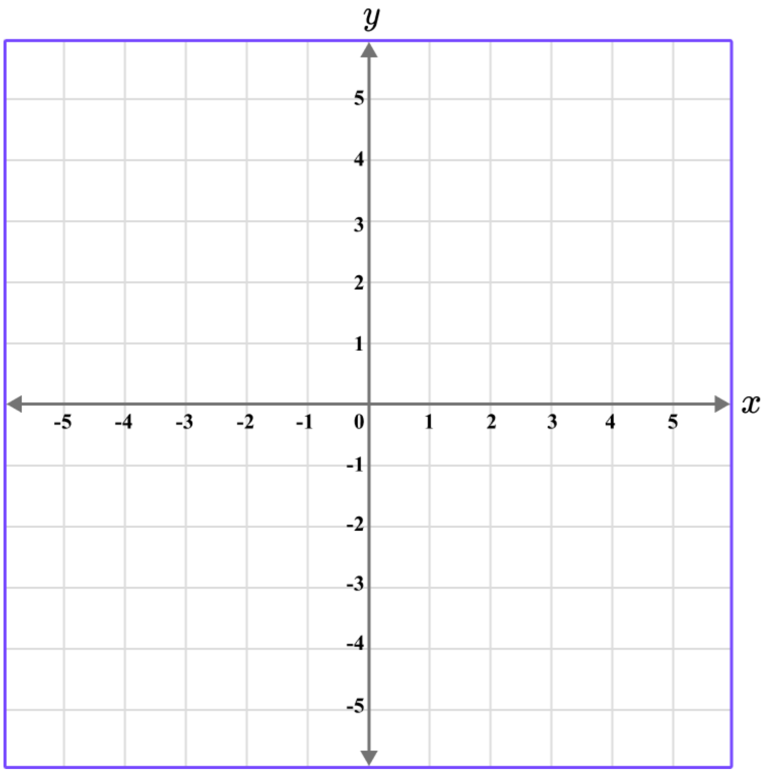
</td><td>

Mentors Note's

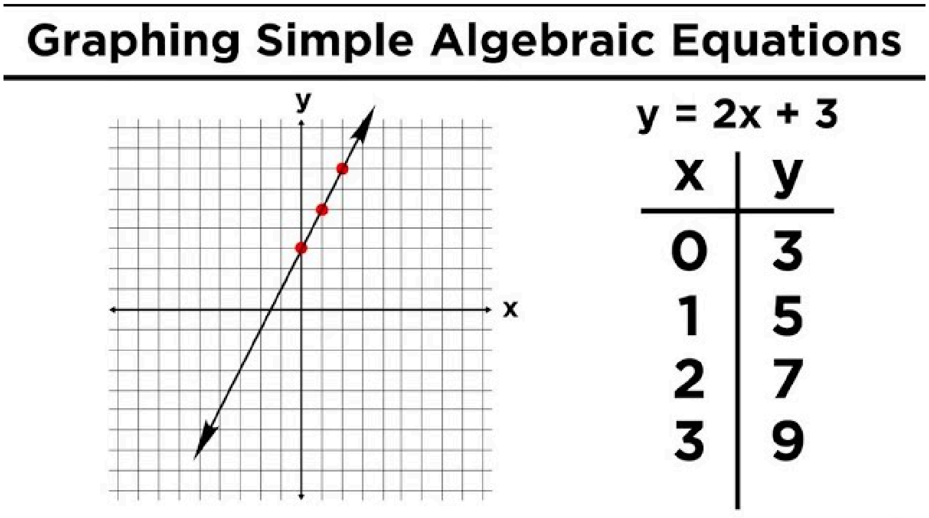
Axis Labels: Never forget to check the scale of the axes. Sometimes one square represents 1 unit, but other times it

</td></tr>
</table>

- The first number **(x-value)** always represents horizontal movement. Positive means move right, and negative means move left. - The second number **(y-value)** always represents vertical movement. Positive means move up, and negative means move down. - The coordinate plane has **four quadrants**, and they are named using Roman numerals going counterclockwise starting from the top-right. **Quadrant I** (1): top-right - x is positive, y is positive - example: (3, 4) **Quadrant II** (2): top-left - x is negative, y is positive - example: (−3, 4) **Quadrant III** (3): bottom-left - x is negative, y is negative - example: (−3, −4) **Quadrant IV** (4): bottom-right - x is positive, y is negative - example: (3, −4) - Example: (3, −4) means go right 3 units, then down 4 units.	***might represent 5 or 10.***



3.2 - Linear Graphs • A **linear graph** is a visual representation of a rule. If you have a rule like y = 2x + 1, every point on that line must follow that rule. • **Table of Values:** To draw a line, pick three values for x (like -1, 0, 1), calculate the y for each, and connect the dots. • Example (Table): For y = 3x - 2, if x = 0, y = -2. If x = 2, y = 4. Connecting (0, -2) and (2, 4) creates the line.	



3.3 - Gradient of Linear Graphs

- The **gradient (m)** measures how **steep** a line is.
- The Formula: Gradient = (change in y) / (change in x). Or: **(y2 - y1) / (x2 - x1).**
- **Positive vs. Negative:** A line going "uphill" from left to right has a positive gradient. A line going "downhill" has a negative gradient.
- Example (Calculation): For a line passing through (1, 2) and (3, 8), the gradient is (8 - 2) / (3 - 1) = 6 / 2 = 3.

Mentors Note's

Order Matters: It doesn't matter which point you call "Point 1," as long as you are consistent. If you start with y2 on top, you must start with x2 on the bottom.

3.4 - Length and Midpoint

- Once you have two points, you can find the space

Mentors Note's

Visualize the Triangle: If you forget the

between them and the exact middle. • **Midpoint:** This is the **average of the coordinates**. Add the x's and divide by 2, then add the y's and divide by 2. • **Length**: Use the **distance formula**, which is just Pythagoras: Square root of [(x2 - x1) squared + (y2 - y1) squared]. • Example (Midpoint): The midpoint of (2, 4) and (6, 10) is [(2+6)/2, (4+10)/2], which is (4, 7). • Example (Length): The length between (0, 0) and (3, 4) is the square root of (3 squared + 4 squared), which is 5.	***length formula, just draw a right-angled triangle on your screen using the line as the long side (hypotenuse). Count the squares for the base and height, then use a squared + b squared = c squared.***

### 3.5 - Equations of Linear Graphs • The standard form for a line is **y = mx + c.** • m is the Gradient: It tells you the **slope**. • **c is the y-intercept**: It tells you where the line crosses the vertical y-axis. • Example (Finding the Equation): If a line has a gradient of 2 and passes through (0, 5), its equation is y = 2x + 5. • Example (Rearranging): If you see 2y = 4x + 6, divide everything by 2 to get y = 2x + 3. Now you can see m = 2 and c = 3.	### Mentors Note's ***The y-intercept trick: If a point has an x-value of 0, like (0, -3), the y-value is automatically your "c". You don't need to do any extra math to find it.***

Mentors’ Sources:

- https://znotes.org/caie/igcse/mathematics-0580/
- https://www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf
- https://www.physicsandmathstutor.com/maths-revision/